

CHAPTER 1

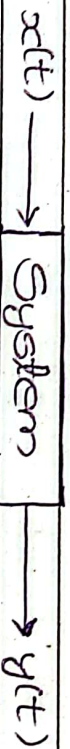
SIGNALS, SYSTEMS AND SIGNAL PROCESSING

Page:	
Date:	/ /

A signal is a function of one or more independent variables that conveys information. In electrical sense, the signal can be voltage or current.

Examples of signal are radio signal, TV. signal, etc

A system is a device which takes signal as i/p and processes on it to give a required op.



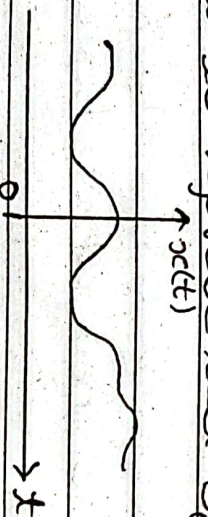
Mathematically, the functional relationship between input and output can be written as :

$$y(t) = F[x(t)]$$

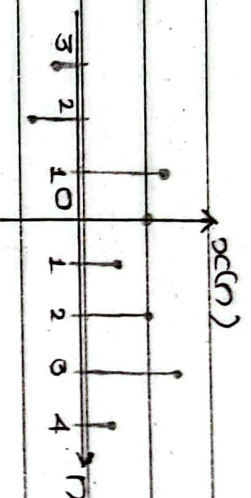
Classification of signals-

Based upon the nature and characteristics in the time domain, the signals may be broadly classified into -

1. Continuous-time signals- A continuous time signal is defined as a signal which varies continuously in time domain. For continuous-time signal, the independent variable is time t . A continuous time signal is represented by $x(t)$.



2. Discrete-time signals - A discrete-time signal is defined only at certain time-instants. For discrete-time signal, the independent variable is time (n) and such signal is represented as $x(n)$.



We may further classify both continuous and discrete time signals as:

1. Deterministic and non-deterministic signals -

The deterministic signal is a signal which can be completely specified in time.

eg- $x(t) = bt$ is a ramp signal whose amplitude increases linearly with time and slope is b .

On the other hand, a non-deterministic signal is one whose occurrence is always random in nature. The pattern of such signal is irregular and are called random signals. eg- thermal noise generated in an electric circuit.

2. Periodic and aperiodic signals - A periodic signal is a type of signal which have fixed patterns

and repeats over and over with some time period 'T'. A signal is said to be periodic if it satisfies the following conditions -

$$x(t) = x(t+T) \text{ for all } t$$

$$x(n) = x(n+N) \text{ for all } n$$

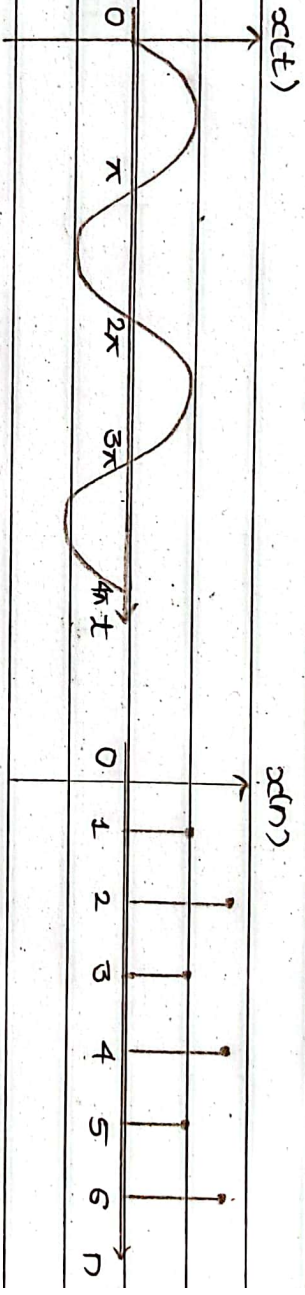


fig- Continuous-time periodic signal

fig- Discrete-time periodic signal

Aperiodic signal is a type of signal which does not repeat after certain interval of time. For aperiodic signal,

$$x(t) \neq x(t+T) \text{ for all } t$$

$$x(n) \neq x(n+N) \text{ for all } n$$

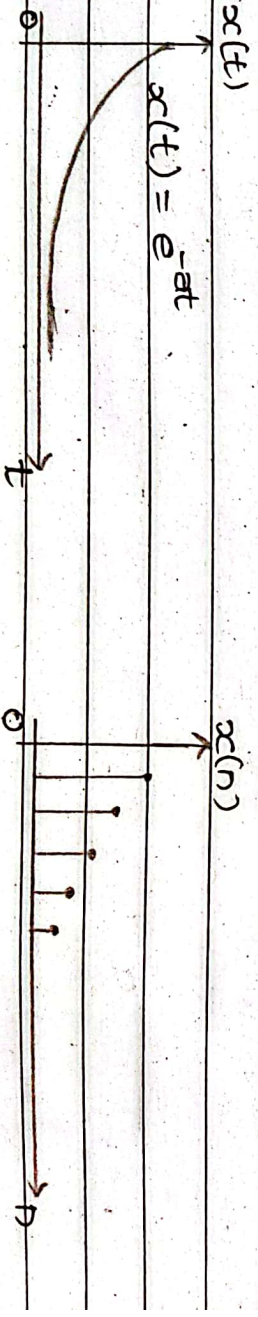


fig- Continuous-time aperiodic signal

fig- Discrete-time aperiodic signal

3. Even and odd signals - An even signal is a type of signal which exhibits symmetry in the time domain. An even signal must satisfy -
 $x(t) = x(-t)$ for continuous time-signal
 $x(n) = x(-n)$ for discrete time signal

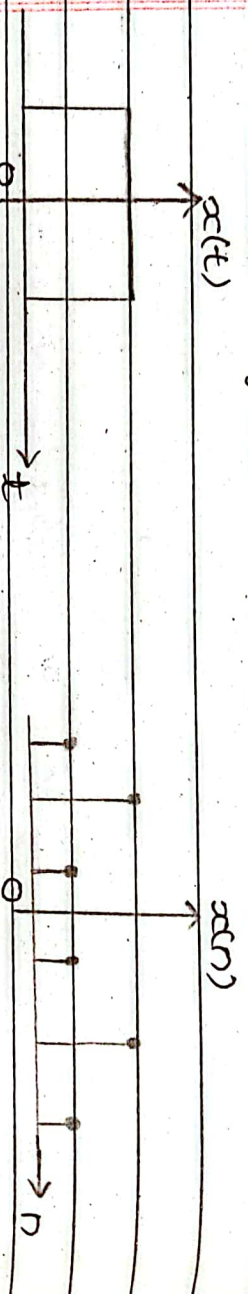


fig- Continuous-time even signal fig- Discrete-time even signal

An odd signal is a type of signal which exhibits anti-symmetry. An odd signal must satisfy the following conditions -

$x(t) = -x(-t)$ for continuous-time signal
 $x(n) = -x(-n)$ for discrete-time signal

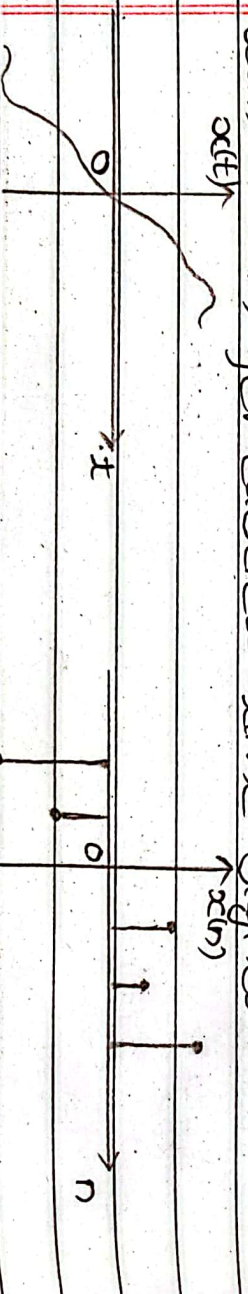
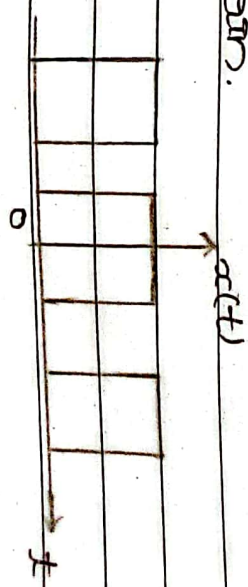
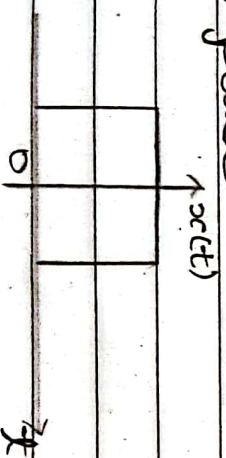


fig- Continuous-time odd signal fig- Discrete-time odd signal

4. Energy and power signals - Energy signal is one which has finite energy and zero average power. On the other hand, power signal is one which has finite average power and infinite energy.

Differences between energy and power signal -

Energy signal	Power signal
1. Total normalized energy is finite and non-zero.	1. The normalized average power is finite and non-zero.
2. The energy is obtained by: $E = \int_{-\infty}^{\infty} x(t) ^2 dt$	2. The average power is obtained by: $P = \frac{1}{T} \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} x(t) ^2 dt$
3. Non-periodic signals are energy signals.	3. Periodic signals are power signals.
4. Power of energy signal is zero.	4. Energy of power signal is infinite.
5. These signals are time limited.	5. These signals exist over infinite time.
6. eg - a single rectangular pulse.	6. eg - a periodic pulse train.



Signal processing systems -

can be classified depending on the type of signal to be processed as -

1. Continuous-time systems - Continuous-time systems are those systems for which both input and output are continuous-time signals. $H(s)$ is the transfer function of continuous time systems. An example of continuous-time system is an analog filter which is used to reduce the noise corrupting a message signal.

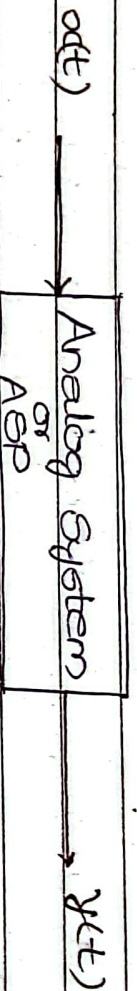


fig-Block diagram of continuous time system

2. Discrete-time systems - Discrete-time systems are those for which both input and output signals are discrete-time signals. $H(z)$ is the transfer function of a discrete-time system. An example of discrete-time system is a digital computer.

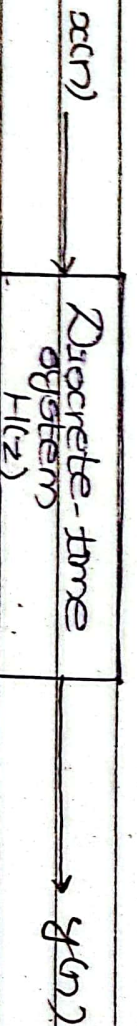


fig-Block diagram of discrete-time system

Signal analysis - is the process of examining & interpreting a signal to understand its properties. It helps to extract information without changing signal. eg - Analyzing an ECG signal to detect heart rate and abnormalities.

Signal processing - is the technique of modifying, transforming or manipulating a signal to improve its quality, extract information or make it suitable for a specific condition.

Digital signal processing - is the numerical representation manipulation of signals, using algorithms implemented on digital computers to perform functions such as filtering, enhancement, compression and analysis.

Basic elements of DSP -

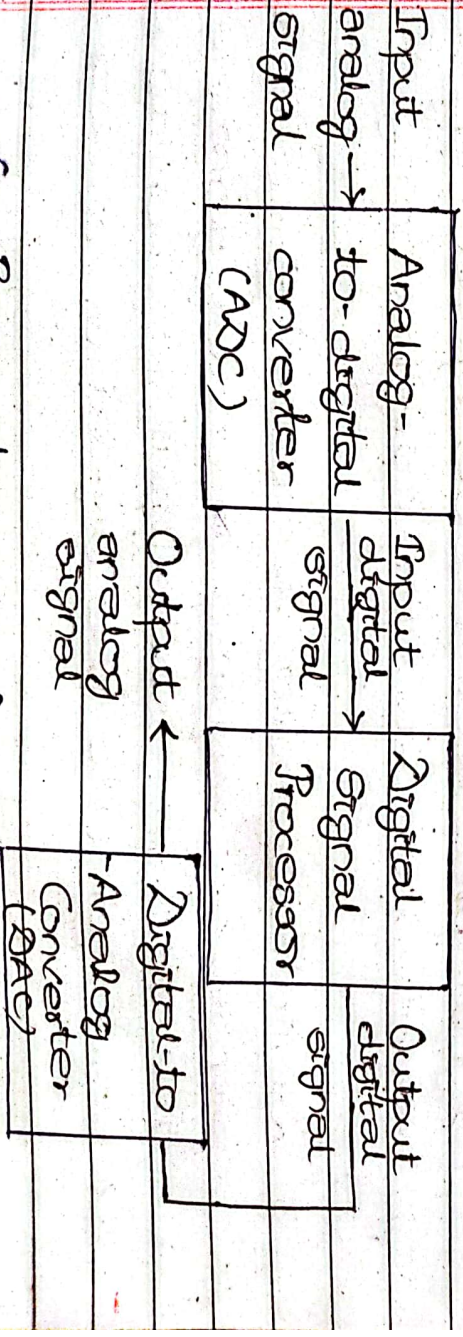


fig - Basic elements of DSP

The above figure shows the basic elements of digital signal processing system. Most of the signals generated are analog in nature and must be digitized to be processed by a DSP system. Hence, input is given through ADC and output is obtained through DAC.

A DSP system consists of following elements

1. Analog-to-digital converter-

The AD converter converts analog input to digital input. This signal is processed by DSP system. The AD converter determines sampling rate and quantization error in digitizing operation.

2. Digital signal processor - It performs amplification, attenuation, filtering, spectral analysis etc. operations on digital data. The digital signal processor consists of ALU, shifter, serial ports, etc. for its functioning.

3. Digital-to-analog converter - Some of the signals are required in analog form. Hence the DSP processor output is given to DAC, which converts the digital output of DSP processor to analog.

cessor into its analog equivalent.

Applications of DSP -

Page :	
Date : / /	

$$\begin{cases} -\infty < n < \infty \\ -\infty < F < \infty \end{cases}$$

Page:
 Date: / /

Continuous-time signal	Discrete-time signal
1. $x_a(t) = A \cos(2\pi Ft + \theta)$ $= A \cos(\Omega t + \theta)$	1. $x(n) = A \cos(2\pi fn + \theta)$ $= A \cos\left(\frac{2\pi n}{T} + \theta\right)$ $= x_a(nT)$
cohere, $\Omega = 2\pi F$ rad/sec $F \rightarrow$ frequency of analog signal in Hz.	cohere, $\omega = 2\pi f$ rad/samples $f \rightarrow$ frequency of digital signal in cycles/sec.
2. $\omega = 2\pi T, f = \frac{F}{F_s}$	2. $\Omega = \frac{\omega}{T}, F = f \cdot T_s$
cohere, $F_s \rightarrow$ sampling frequency	$-\frac{\pi}{T} \leq \Omega \leq \frac{\pi}{T}$ $-\frac{F}{2} \leq F < \frac{F}{2}$

Some terms related to sampling of analog signal

1. Sampling frequency, F_s : It is the number of samples per second while converting continuous time signals into discrete-time signals.

$$F_s = \frac{\text{Bit rate}}{\text{Bits/sample (b)}}$$

$$\text{cohere, } b = \log_2(L) \text{ bits per sample}$$

2. Nyquist criteria, $F_s \geq 2F_{\max}$: It is the criteria that has to be fulfilled for the reconstruction of signal from discrete-time signal.

Here, F_{max} is the maximum frequency contained in continuous-time signal.

3. Nyquist rate, $f_N = 2F_{max}$: It is the minimum sampling frequency required for the reconstruction of the signal.

4. Folding frequency (or Nyquist frequency), $f_N/2$: The highest frequency that can be reconstructed or measured using discretely sampled data.

Nyquist interval = $\frac{1}{2f_{max}}$

Resolution (Δ) = $\frac{x_{max} - x_{min}}{1 \rightarrow \text{no of quantization levels}}$

$x_{max} = A_1 + A_2$

$x_{min} = -x_{max}$

To convert any continuous signal into discrete, $t = nT$ or $\frac{T}{f_s}$

Q.1. Digital communication link carries binary coded words representing samples of ip signal $x_g(t) = 3\cos 600\pi t + 2\cos 800\pi t$.

iv) Resolution

The link is operated at 10,000 bits/sec. Each ip sample is quantized into 1024 different voltage levels. Determine

i) Nyquist rate for $x_a(t)$

ii) Sampling frequency and folding frequency.

iii) frequencies in resulting discrete-time signal.

sol: The given input signal is :

$$x_a(t) = 3 \cos 600\pi t + 2 \cos 800\pi t$$

i) Comparing with $x_a(t) = A_1 \cos 2\pi f_1 t + A_2 \cos 2\pi f_2 t$,
 $A_1 = 3$, $A_2 = 2$, $f_1 = 300 \text{ Hz}$ and $f_2 = 400 \text{ Hz}$.

$$\therefore f_{\max} = 400 \text{ Hz}.$$

$$\therefore \text{Nyquist rate, } f_N = 2 f_{\max} \\ = 2 \times 400$$

$$\therefore f_N = 800 \text{ Hz}.$$

ii) Sampling frequency, $f_s = \frac{\text{Bit rate}}{\text{Bits/Sample (b)}}$

Here, Bit rate = 10000 bits/sec.

And, No. of quantization level (L) = 1024

$$\therefore b = \log_2 (1024) = 10 \text{ bits/sample}.$$

$$f_s = \frac{10000}{10} = 1000 \text{ samples/sec}.$$

iii) For $f_s = 1000 \text{ Hz}$,

$$x(n) \times x(nT) = x\left(\frac{n}{f_s}\right) = 3 \cos 600\pi \left(\frac{n}{1000}\right) + 2 \cos 800\pi \left(\frac{n}{1000}\right)$$

$$\therefore x(n) = 3 \cos 0.6\pi n + 2 \cos 0.8\pi n$$

Comparing with $x(n) = A_1 \cos 2\pi f_1 n + A_2 \cos 2\pi f_2 n$,
 $f_1 = 0.3$ and $f_2 = 0.4$.

iv) We have, resolution $(\Delta) = \frac{x_{max} - x_{min}}{L-1}$

$$= \frac{2(A_1 + A_2)}{L-1} = \frac{2(3+2)}{1023}$$

$$\therefore \Delta = 0.009775 \approx 9.775 \text{ mV}$$

2. Consider an analog signal $x_a(t) = 10 \cos 100\pi t$.

i) Determine the minimum sampling rate to avoid aliasing.

ii) Suppose that the signal is sampled at $f_s = 75 \text{ Hz}$. What is the discrete-time signal obtained after sampling?

iii) Find discrete-time signal sampled at $f_s = 75 \text{ Hz}$.

iv) What is the folding frequency for step (ii).

sol: Given, $x_a(t) = 10 \cos 100\pi t$.

Comparing with $x_a(t) = A \cos 2\pi f t$, we have

$$F = 500 \text{ Hz}$$

$$\therefore f_{max} = 500 \text{ Hz}$$

The criteria for minimum sampling rate to avoid aliasing is $f_s \geq 2f_{max}$

$$\therefore f_s \geq 100 \text{ Hz.}$$

ii) For $f_s = 200 \text{ Hz}$,

$$x(n) \cong x(nT) = x\left(\frac{n}{f_s}\right)$$

$$= 10 \cos 100\pi \left(\frac{n}{200}\right)$$

$$\therefore x(n) = 10 \cos 0.5\pi n$$

iii) For $f_s = 75 \text{ Hz}$,

$$x(n) \cong x(nT) = x\left(\frac{n}{f_s}\right)$$

$$= 10 \cos 100\pi \left(\frac{n}{75}\right)$$

$$\therefore x(n) = 10 \cos 1.33\pi n.$$

x7) Folding frequency for (ii) $= \frac{f_s}{2} = \frac{200}{2} = 100 \text{ Hz.}$

3. Consider a signal $x_a(t) = 10 \cos 2000\pi t + 5 \cos 1000\pi t$ is said to be sampled. Determine

i) Nyquist rate and Nyquist interval

ii) If the signal is sampled at 4 kHz. Will the signal be recovered from its sample?

Q8: Given, $x_a(t) = 10 \cos 2000\pi t + 5 \cos 10000\pi t$

∴ Nyquist rate, $F_N = 2F_{max}$

$$\therefore F_N =$$

Here, $F_1 = 1000 \text{ Hz}$ and $F_2 = 5000 \text{ Hz}$

$$\therefore F_{max} = 5000 \text{ Hz}$$

So, $F_N = 10000 \text{ Hz}$.

And, Nyquist interval, $\frac{1}{2F_{max}} = 10^{-4} \text{ Hz sec.}$

∴ Given, Sampling frequency, $F_s = 4 \text{ kHz}$.
Also,

$$x(n) \equiv x(nT) = x\left(\frac{n}{F_s}\right)$$

$$= 10 \cos 2000\pi \left(\frac{n}{4000}\right) + 5 \cos 10000\pi \left(\frac{n}{4000}\right)$$

$$= 10 \cos 0.5\pi \left(\frac{1}{4}\right)n + 5 \cos 2.5\pi \left(\frac{5}{4}\right)n$$

$$= 10 \cos 2\pi \left(\frac{1}{4}\right)n + 5 \cos 2\pi \left(1 + \frac{1}{4}\right)n$$

$$= 10 \cos 2\pi \left(\frac{1}{4}\right)n + 5 \cos \left(2\pi + \frac{2\pi}{4}\right)n$$

$$= 10 \cos 2\pi \left(\frac{1}{4}\right)n + 5 \cos 2\pi \left(\frac{1}{4}\right)n$$

$$\therefore x(n) = 15 \cos 2\pi \left(\frac{1}{4}\right)n.$$

Since, $\therefore x(t) = 15 \cos 2\pi (1000)t$.

We observed that the reconstructed signal contains only one signal with frequency 1000Hz and amplitude 15. This shows that with the sampling rate of 4000 Hz, the original signal is not recovered from its sample.

OR To reconstruct a signal after sampling, the sampling frequency must be greater than twice the highest frequency in the signal i.e.

$$f_s \geq 2f_{\max}$$

Here, $f_{\max} = 5\text{KHz}$. So, $f_s \geq 10\text{KHz}$.

But we are sampling at 4KHz which is below Nyquist rate. So, the signal cannot be perfectly sampled recovered from its samples if sampled at 4KHz.

Need of DSP over ASP-

Disadvantages of DSP -

1. DSP systems are expensive for small applications.
2. DSP system, especially real-time ones, consume a lot of energy which is not ideal for battery powered devices.

3. If the signal is not sampled properly, then imp-

important details can be lost. This is called aliasing.

4. Real-world signals are analog, so we need ADC before DSP can work and DAC after, which adds extra hardware and possible signal degradation.

5. DSP systems are complex in design.

Sampling continuous signals and spectral properties of sampled signals -

Sampling - is the process of converting continuous time signal into a discrete time signal by taking samples of continuous time signals at discrete time instants.

If $x_a(t)$ is the input to the sampler, then the output is obtained as:

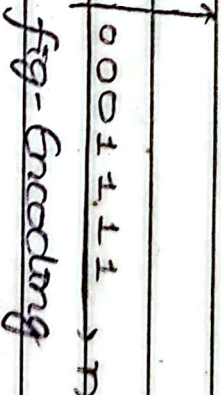
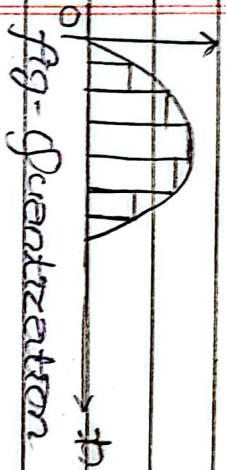
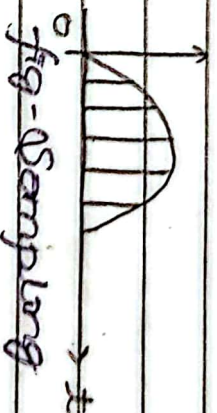
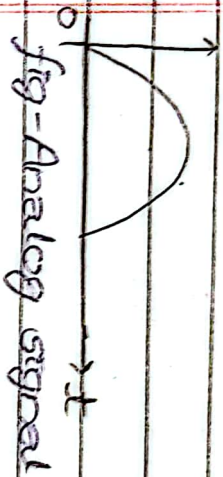
$$x_s(n) = x_a(nT)$$

where, $T \rightarrow$ sampling interval

Quantization - is the process of approximating the amplitude of each sample to the nearest value from a finite set of levels. It converts infinite range analog values into a finite set of digital values. Rounding and truncation are typical examples of quantization. This introduces

quantization error.

Encoding - is the process of converting each quantized sample into a binary number so it can be processed, stored or transmitted digitally.



Sampling of analog signals -

Let us consider an analog signal.

$$x_a(t) = x_a(nT), \quad -\infty < n < \infty$$

where, $x_a(t)$ = i/p signal

T = sampling interval

$\frac{1}{T}$ = sampling frequency (f_s)

In discrete time, $t = nT = \frac{n}{f_s}$

$$\text{Let } x_a(t) = A \cos(2\pi f t + \theta)$$

Replacing t by nT , we have,

$$x_a(nT) = A \cos\left(2\pi f \frac{n}{f_s} + \theta\right)$$

⇒ $x_s(nT) \approx x_a(nT) = A \cos\left(2\pi \frac{f}{f_s} n + \theta\right)$

If we compare with discrete-time sinusoid,

we get $f = \frac{F}{f_s}$ or $\omega = \Omega T$

It is called relative or normalized frequency.

Aliasing effect - Aliasing effect is the phenomenon in DSP where a continuous signal is sampled at a rate lower than twice its highest frequency causing different frequency components to become indistinguishable in sampled signal. This results in distorted or misleading representations of original signal in digital domain.

CHAPTER 2

DISCRETE-TIME SIGNALS & SYSTEMS

Page :
Date : 11 /

Digital signals are discrete in both time & amplitude. Signals that are discrete in time but continuous in amplitude are discrete-time signals.

A discrete-time signal is a function of independent variable which is an integer. A discrete time signal can be represented as:

i) Functional representation -

$$x(n) = \begin{cases} 1, & \text{for } n=1, 2, 3 \\ 4, & \text{for } n=2 \\ 0, & \text{otherwise} \end{cases}$$

ii) Tabular representation -

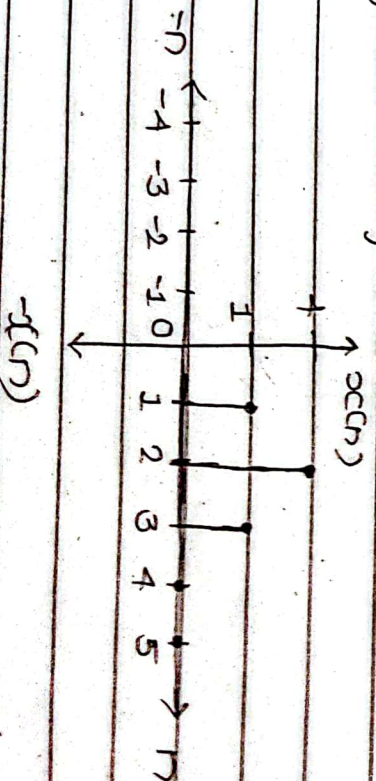
$x(n)$	+1	0	1	2	3
n	-1	0	1	2	3

iii) Sequential representation -

$$x(n) = \{0, 1, 4, 1, 0, 0\}$$

'↑' represents the position of $n=0$.

iv) Graphical representation -

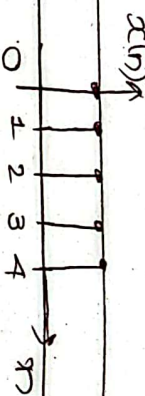


* Elementary signals are those signals that can be used as building blocks to build more complex signals.

Some elementary D.T. signals -

Page: / /
Date: / /

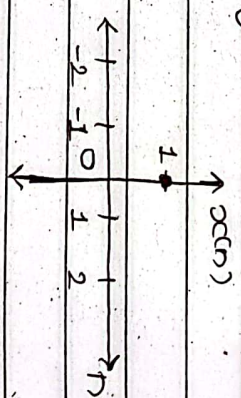
1. Unit-step signal - The unit-step function is that type of elementary function which exists unity for positive side and zero for negative side. It is denoted by $u(t)$ & is given by,

$$u(t) = \begin{cases} 1 & ; n \geq 0 \\ 0 & ; \text{otherwise} \end{cases}$$


Sequential representation:

$$u(n) = \{ \dots, 0, 0, \underset{\uparrow}{1}, 1, 1, \dots \}$$

2. Unit-delta/Impulse signal - is a type of elementary signal whose magnitude is unity only at the origin. It is denoted by $\delta(t)$ and is given by,

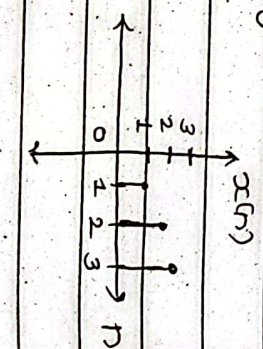
$$\delta(n) = \begin{cases} 1 & ; n = 0 \\ 0 & ; \text{otherwise} \end{cases}$$


$$\delta(n) = \{ \dots, 0, 0, 1, 0, 0, \dots \}$$

It is also known as unit sample signal.

3. Ramp signal - It is denoted by $r(n)$ or $u_r(n)$ & mathematically given by,

$$u_r(n) = \begin{cases} n & ; n \geq 0 \\ 0 & ; \text{otherwise} \end{cases}$$

$$r(n) = \{ \underset{\uparrow}{0}, 1, 2, 3, \dots \}$$


4. Exponential signal - It is given by :

$$x(n) = a^n \text{ for all } n.$$

where, $a = re^{j\theta}$.

If 'a' is real, then $x(n)$ is also real. If 'a' is complex, then $a = re^{j\theta}$.

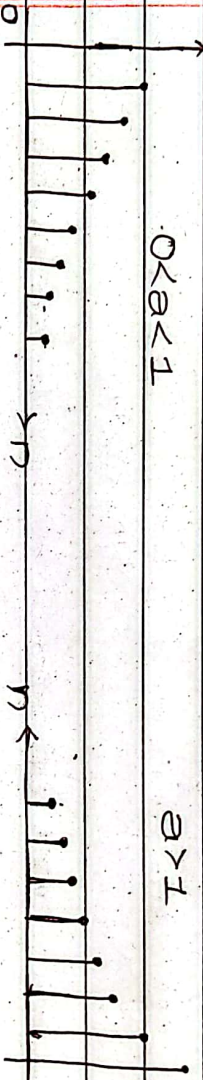
$$x(n) = re^{j\theta} = r(\cos\theta + j\sin\theta)$$

$$\therefore x(n) = r\cos\theta + j r\sin\theta.$$

$$x_r(n) = r\cos\theta \rightarrow \text{real part}$$

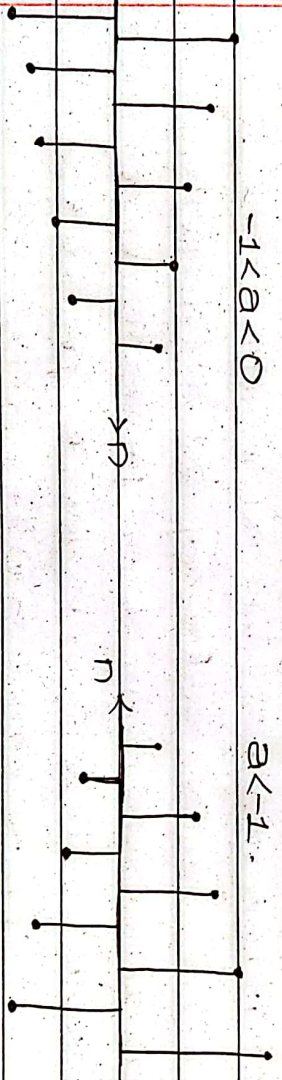
$$x_i(n) = r\sin\theta \rightarrow \text{imaginary part}$$

$x(n)$



$0 < a < 1$

$a > 1$



$-1 < a < 0$

$a < -1$

Classification of discrete-time signal -

1. Energy signal and power signal - The energy of a signal can be finite or infinite. If the energy of a signal is finite, then it is known as energy signal.

Many signals that possess infinite energy have a finite average power. The average power of a discrete-time signal $x(n)$ is defined as -

$$P = \frac{1}{2N+1} \lim_{N \rightarrow \infty} \sum_{n=-N}^N |x(n)|^2$$

If we define the energy signal $x(n)$ over the finite interval $-N \leq n \leq N$.

$$E_N = \sum_{n=-N}^N |x(n)|^2$$

Then, the energy signal E is given by,

$$E = \lim_{N \rightarrow \infty} E_N \text{ and } P = \lim_{N \rightarrow \infty} \left(\frac{1}{2N+1} \right) E_N$$

2. Periodic and aperiodic signal - A signal $x(n)$ is periodic with period N ($N > 0$) if and only if $x(n+N) = x(n)$ for all N . --- (1)

If there is no value of ' N ' that satisfies eq (1) then the signal is non-periodic or aperiodic.

A discrete-time sinusoid is periodic only if ' f ' is a fractional number.

Proof:

From the condition of periodicity:

$$x(n+N) = x(n) \text{ for all } N.$$

Let us consider a sinusoid with frequency f_0 .

$$x(n) = A \cos(2\pi f_0 n + \theta)$$

$$\& x(n+N) = A \cos[2\pi f_0 (n+N) + \theta]$$

For a sinusoid with frequency ' f_0 ' is said to be periodic if $x(n+N) = x(n)$.

$$\text{i.e. } A \cos[2\pi f_0 n + \theta] = A \cos[2\pi f_0 (n+N) + \theta]$$

$$\Rightarrow A \cos[2\pi f_0 n + 2\pi f_0 N + \theta] = A \cos[2\pi f_0 n + \theta]$$

which is true if and only if there exists an integer ' k ' such that: ~~$k = 2\pi f_0 N = 2\pi k$~~

$$\therefore f_0 = \frac{k}{N}, \text{ where } k=0,1,2,\dots$$

$\therefore f_0$ must be a rational number.

3. Even (symmetric) or odd (asymmetric) signal.

Properties of system-

1. Static and dynamic - A discrete-time signal is called static or memory-less if its output at any instant ' n ' depends on present x_p but not on past or future x_p . eg- $y(n) = 2x(n) - x^2(n)$.

On the other hand, a D.T signal is called dynamic or memory-less if its x_p at any instant ' n ' depends present x_p as well as past or future.

$$\text{eg- } y(n) = ax(n-1) + x^2(n).$$

2. Time-variant and time-invariant - A system is time-invariant if its x_p - x_p characteristics do not change with time.

3. Linearity - A linear system is one that satisfies superposition principle. A system is linear if & only if, $T[a_1 x_1(n) + a_2 x_2(n)] = a_1 T[x_1(n)] + a_2 T[x_2(n)]$ or, $a_1 x_1(n) + a_2 x_2(n) = a_1 y_1(n) + a_2 y_2(n)$.
If the system is non-linear, then it does not satisfy superposition principle.

4. Causality - A system is said to be causal if the o/p of the system $y(n)$ at any time 'n' depends only on present and past i/p $[x(n), x(n-1), \dots]$ but does not depend on future values $[x(n+1), x(n+2), \dots]$. Causal system is memory-less system.

Mathematically,

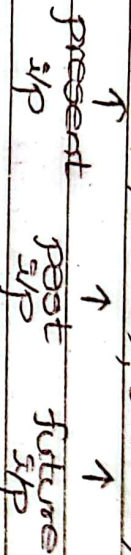
$$y(n) = F[x(n), x(n-1), \dots]$$

A system is said to be non-causal & anti

causal system if its o/p depends not only on present or past i/p but also on future i/p.

Mathematically,

$$y(n) = F[x(n), x(n-1), x(n+1), x(n+2)]$$



5. Stability - To define the stability of the system, we use term 'BIBO' i.e. Bounded Input Bounded Output. An initially relaxed system is BIBO stable if and only if every bounded i/p produces a bounded o/p.

Input is bounded i.e. $|x(n)| < M_x < \infty$.

Output is bounded i.e. $|y(n)| < M_y < \infty$

where, M_x and M_y are finite numbers whose values are less than infinity.

An initially relaxed system is BIBO unstable if and only if bounded i/p does not produce bounded o/p.

Recursive and non-recursive DTIS - A system whose o/p at any time ' n ' depends on any no. of past values of o/p $y(n-1), y(n-2) \dots$ is called recursive system i.e. $y(n) = F[y(n-1), y(n-2) \dots]$

A system whose op at any time 'n' depends on present and past ip, then it's called non-recursive system.

$$\text{i.e } y(n) = F[x(n), x(n-1), x(n-2), \dots, x(n-m)]$$

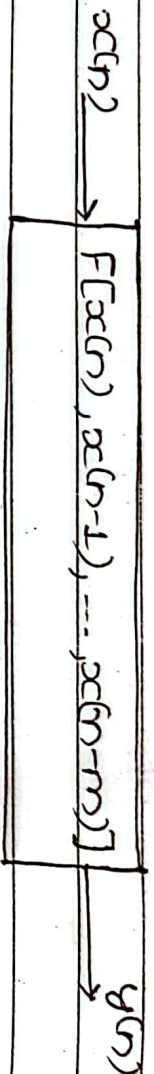


fig-Non-recursive DTS

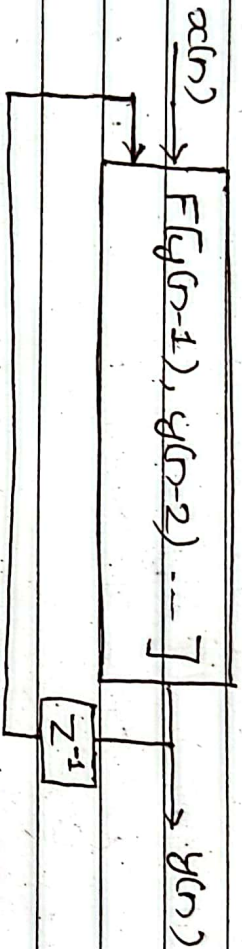
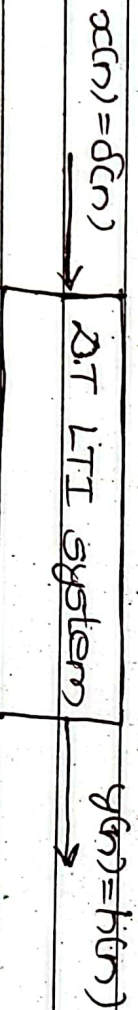


fig-Recursive DTS

Linear Time Invariant system- If the system has both linearity and time-invariance properties, then the system is said to be LTI system.



Here, $\delta(n)$ is the unit-impulse ip ⁱⁿ LTI system. $h(n)$ is the unit-impulse response of LTI system.

$\therefore$ Any LTI system can be completely chara-

denoted in terms of unit-impulse response.

Convolution sum - Let us consider relaxed LTI system. A relaxed system means if i/p $x(n)$ is zero, then o/p $y(n)$ is also zero.

Let us say unit-impulse function $\delta(n)$ is applied to the system, then its o/p is denoted by $h(n)$. Here, $h(n)$ is called an impulse response of the system.

$$i) \delta(n) \xrightarrow{T} y(n) = h(n)$$

The system is time-invariant. If we delay i/p by 'k' samples, then the o/p should be delayed by the same amount 'k'.

$$ii) \delta(n-k) \xrightarrow{T} y(n) = h(n-k)$$

Also, multiplying both sides by $x(k)$, we get,

$$iii) x(k) \delta(n-k) \xrightarrow{T} y(n) = x(k) \cdot h(n-k)$$

Since the system is linear, we use superposition principle. Hence summation on both sides, we

$$\text{get,} \quad \sum_{k=-\infty}^{\infty} x(k) \delta(n-k) \xrightarrow{T} y(n) = \sum_{k=-\infty}^{\infty} x(k) \cdot h(n-k)$$

$$\therefore \sum_{k=-\infty}^{\infty} x(k) \cdot \delta(n-k) = \sum_{k=-\infty}^{\infty} x(k) \cdot h(n-k) = y(n)$$

$$\therefore y(n) = \sum_{k=-\infty}^{\infty} x(k) \cdot h(n-k)$$

∴ Required convolution of $x(n)$ & $h(n)$ is given by,
$$x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k) \cdot h(n-k)$$

Q.1. Find the convolution sum of two sequences:

$x(n) = \{2, 4, 6\}$ and $h(n) = \{3, 0, 1\}$.

∴ Given, $x(n) = \{2, 4, 6\}$

$h(n) = \{3, 0, 1\}$

∴ $x(0) = 2$, $x(1) = 4$, $x(2) = 6$.

$h(0) = 3$, $h(1) = 0$, $h(2) = 1$.

The convolution sum of $x(n)$ and $h(n)$ is given by
$$x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k) \cdot h(n-k) \quad \dots (1)$$

Total length of $y(n)$ is given by the sum of length of $x(n)$, $h(n)$ and '1' is subtracted (since origin is same for both signal i.e. origin is repeated).

∴ Length of $x(n) = L = 3$

Length of $h(n) = M = 3$

∴ Length of $y(n) = L + M - 1 = 3 + 3 - 1 = 5$.

To determine the upper and lower limit of 'k':

Lower limit of $y(n) = \text{lower limit of } x(n) + \text{lower}$

limit of $h(n) = 0 + 0 = 0$.

Upper +

Upper limit of $y(n) = \text{Upper limit of } x(n) + \text{Upper limit of } h(n) = 2+2=4.$

Eg¹ (1) becomes,

$$y(n) = \sum_{k=0}^4 x(k) \cdot h(n-k) \quad \dots (2)$$

At $n=0$: $y(0) = \sum_{k=0}^4 x(k) \cdot h(0-k)$

$$= x(0) \cdot h(0) + x(1) \cdot h(-1) + x(2) \cdot h(-2) + x(3) \cdot h(-3) + x(4) \cdot h(-4)$$

$$= 2 \times 3 + 4 \times 0 + 0 + 0 + 0$$

$$\therefore y(0) = 6.$$

At $n=1$: $y(1) = \sum_{k=0}^4 x(k) \cdot h(1-k)$

$$= x(0) \cdot h(1) + x(1) \cdot h(0) + x(2) \cdot h(-1)$$

$$= 2 \times 0 + 4 \times 3$$

$$\therefore y(1) = 12.$$

At $n=2$: $y(2) = \sum_{k=0}^4 x(k) \cdot h(2-k)$

$$= x(0) \cdot h(2) + x(1) \cdot h(1) + x(2) \cdot h(0)$$

$$= 2 \times 1 + 4 \times 0 + 6 \times 3$$

$$\therefore y(2) = 20$$

At $n=3$: $y(3) = \sum_{k=0}^4 x(k) \cdot h(3-k)$

$$= x(0) \cdot h(3) + x(1) \cdot h(2) + x(2) \cdot h(1) + x(3) \cdot h(0)$$

$$\therefore y(3) = 4 \times 1 + 6 \times 0 + 0 = 4.$$

$$\text{At } n=1: y(1) = \sum_{k=0}^1 x(k) \cdot y(1-k)$$

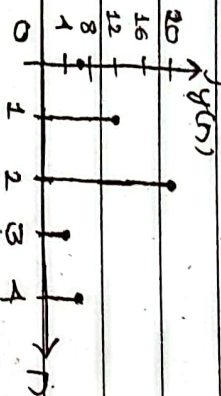
$$= x(0) \cdot h(1) + x(1) \cdot h(0) + x(2) \cdot h(2) + x(3) \cdot h(1)$$

$$= 6 \times 1$$

$$\therefore y(1) = 6$$

$$\therefore y(n) = \{6, 12, 20, 4, 6\}$$

The graphical representation is:



Check:

$$y(n) = \{6, 12, 20, 4, 6\}$$

$$x(n) = \{6, 12, 20, 4, 6\} \quad \therefore y(n) = \{6, 12, 20, 4, 6\}$$

$$4 \{12, 20, 4\}$$

$$6 \{18, 6, 6\}$$

$$R. \text{ If } x_1(n) = \{1, 2, 1\}, x_2(n) = \{2, 3\}$$

$$\text{Sol: Given, } x_1(n) = \{1, 2, 1\}$$

$$x_2(n) = \{2, 3\}$$

$$\text{Here, } x_1(-2) = 1, x_1(-1) = 2, x_1(0) = 1$$

$$x_2(-1) = 2, x_2(0) = 3$$

$$\text{Length of } x_1(n) = L = 3$$

$$\text{Length of } x_2(n) = M = 2$$

$$\therefore \text{Length of } y(n) = L + M - 1 = 3 + 2 - 1 = 4$$

Also, lower limit of $y(n)$ $[k] = -2-1 = -3$
Upper limit of $y(n)$ $[k] = 0+0 = 0$.

The convolution of $x_1(n)$ and $x_2(n)$ is given by:

$$y(n) = x_1(n) * x_2(n) = \sum_{k=-3}^0 x_1(k) \cdot x_2(n-k)$$

At $n = -3$: $y(-3) = \sum_{k=-3}^0 x_1(k) \cdot x_2(-3-k)$

$$= x_1(-3) \cdot x_2(-3+3) + x_1(-2) \cdot x_2(-3+2) + x_1(-1) \cdot x_2(-3+1) + x_1(0) \cdot x_2(-3)$$

$$= 1 \times 2 + 0$$

$$\therefore y(-3) = 2$$

At $n = -2$: $y(-2) = \sum_{k=-3}^0 x_1(k) \cdot x_2(-2-k)$

$$= x_1(-3) \cdot x_2(-2+3) + x_1(-2) \cdot x_2(-2+2) + x_1(-1) \cdot x_2(-2+1) + x_1(0) \cdot x_2(-2-0)$$

$$= 1 \times 3 + 2 \times 2$$

$$\therefore y(-2) = 7$$

At $n = -1$: $y(-1) = \sum_{k=-3}^0 x_1(k) \cdot x_2(-1-k)$

$$= x_1(-3) \cdot x_2(-1+3) + x_1(-2) \cdot x_2(-1+2) + x_1(-1) \cdot x_2(-1+1) + x_1(0) \cdot x_2(-1-0)$$

$$= 2 \times 3 + 1 \times 2$$

$$\therefore y(-1) = 8$$

$$\text{At } n=0: y(0) = \sum_{k=-3}^0 x_1(k) \cdot x_2(b(0-k))$$

$$= x_1(0) \cdot b(0) + x_1(-1) \cdot b(1) + x_1(-2) \cdot b(2) + x_1(-3) \cdot b(3)$$

$$\therefore y(0) = 3$$

$$\therefore y(n) = \{2, 7, 8, 3\}$$

3. Find output $y(n)$ for an LTI system having impulse response $h(n) = \{-1, 0, 3, 1\}$ and $x(n) = \{1, 3, 2, -2\}$.

sol: Given, $h(n) = \{-1, 0, 3, 1\}$ and $x(n) = \{1, 3, 2, -2\}$
 $\therefore x(0) = 1, x(1) = 3, x(2) = 2, x(3) = -2$

$$h(0) = -1, h(1) = 0, h(2) = 3, h(3) = 1$$

Now,

$$\text{Length of } x(n) = L = 4$$

$$\text{Length of } h(n) = M = 4$$

$$\therefore \text{Length of } y(n) = L + M - 1 = 4 + 4 - 1 = 7$$

$$\text{Upper limit of } y(n) = 3 + 3 = 6$$

$$\text{Lower limit of } y(n) = 0 + 0 = 0$$

The convolution of $x(n)$ & $h(n)$ is given by,

$$y(n) = x(n) * h(n) = \sum_{k=0}^6 x(k) \cdot h(n-k) \quad \dots (1)$$

$$\text{At } n=0: y(0) = \sum_{k=0}^6 x(k) \cdot h(-k)$$

$$= x(0) \cdot h(0) + x(1) \cdot h(-1) + x(2) \cdot h(-2)$$

$$= -1.$$

$$\therefore y(0) = -1$$

$$\text{At } n=1: y(1) = \sum_{k=0}^6 x(k) \cdot h(1-k)$$

$$= x(0) \cdot h(1) + x(1) \cdot h(0) + x(2) \cdot h(-1)$$

$$= 1 \times 0 + 3 \times 1$$

$$\therefore y(1) = 3$$

$$\text{At } n=2: y(2) = \sum_{k=0}^6 x(k) \cdot h(2-k)$$

$$= x(0) \cdot h(2) + x(1) \cdot h(1) + x(2) \cdot h(0) + x(3) \cdot h(-1)$$

$$= 1 \times 3 + 3 \times 0 + 2 \times 1$$

$$\therefore y(2) = 1$$

$$\text{At } n=3: y(3) = \sum_{k=0}^6 x(k) \cdot h(3-k)$$

$$= x(0) \cdot h(3) + x(1) \cdot h(2) + x(2) \cdot h(1) + x(3) \cdot h(0)$$

$$= 1 \times 1 + 3 \times 3 + 2 \times 0 + (-2 \times 1)$$

$$\therefore y(3) = 4$$

$$\text{At } n=4: y(4) = \sum_{k=0}^6 x(k) \cdot h(4-k)$$

$$= x(0) \cdot h(4) + x(1) \cdot h(3) + x(2) \cdot h(2) + x(3) \cdot h(1) + x(4) \cdot h(0)$$

$$= 3 \times 1 + 2 \times 3 + (-2 \times 0) + 0$$

$$\therefore y(4) = 9$$

$$\text{At } n=5: y(5) = \sum_{k=0}^6 x(k) \cdot y(5-k)$$

$$= x(0) \cdot h(5) + x(1) \cdot h(4) + x(2) \cdot h(3) + x(3) \cdot h(2) + x(4) \cdot h(1) + x(5) \cdot h(0)$$

$$= 2 \times 1 + (-2) \times 3$$

$$\therefore y(5) = -4.$$

$$\text{At } y=6: y(6) = \sum_{k=0}^6 x(k) \cdot y(6-k)$$

$$= x(0) \cdot h(6) + x(1) \cdot h(5) + x(2) \cdot h(4) + x(3) \cdot h(3) + x(4) \cdot h(2) + x(5) \cdot h(1) + x(6) \cdot h(0)$$

$$= -2 \times 1$$

$$\therefore y(6) = -2$$

$$\therefore y(n) = \{2, -1, -3, 1, 1, 2, 9, -4, -2\}$$

4. $x(n) = a^n$ for $0 \leq n \leq 6$ and $h(n) = 1$ for $0 \leq n \leq 4$.

Sol: Given, $x(n) = a^n$

$$\therefore x(n) = 2^n$$

$$\text{And, } h(n) = 1$$

$$\therefore h(0) = 1, h(1) = 1, h(2) = 1, h(3) = 1, h(4) = 1.$$

Now,

$$\text{Length of } x(n) = L =$$

$$\text{Length of } h(n) = M = 4$$

$$\therefore \text{Length of } y(n) = L + M - 1 = 4 + 1 =$$

5. Find the convolution between two signals where $x(n) = u(n)$ and impulse response $h(n) = a^n u(n)$, $|a| < 1$.

Sol: Given, $x(n) = u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & \text{otherwise} \end{cases}$

$$\therefore x(n) = \begin{cases} 1, & n \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

And,

$$h(n) = \begin{cases} a^n, & n \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\therefore h(n) = \{a^0, a^1, a^2, \dots\}$$

The convolution of $x(n)$ and $h(n)$ is given by,

$$y(n) = x(n) * h(n-k) = \sum_{k=-\infty}^{\infty} x(k) * h(n-k)$$

$$= \{ \dots + x(n)h(n-k) + x(n-1)h(n-k+1) + x(n-2)h(n-k+2) + \dots \}$$

$$= \{ \dots + x(n-2) * h(n+2) + x(n-1)h(n+1) + x(n)h(n) + x(n+1)h(n+1) + \dots \}$$

$$= \{ \dots + 1 + a + a^2 + a^3 + \dots \}$$

$$\text{At } n=0, h(0) = a^0 u(0) = 1$$

$$\text{At } n=1, h(1) = a^1 u(1) = a$$

$$\text{At } n=2, h(2) = a^2 u(2-2) = a^2 \dots \text{ and so on.}$$

$$\therefore y(n) = 1 + a + a^2 + a^3 + \dots + a^n$$

$$\therefore y(n) = \sum_{k=0}^n a^k, \quad n \geq 0.$$

$$\therefore \sum_{k=N_1}^{N_2} a^k = \frac{a^{N_1} - a^{N_2+1}}{1-a} ; a \neq 1$$

$$\therefore y(n) = \frac{a^0 - a^{n+1}}{1-a} ; a \neq 1 \text{ \& } n \geq 0$$

$$\therefore y(n) = \begin{cases} \frac{1-a^{n+1}}{1-a} & ; a \neq 1 \text{ \& } n \geq 0 \\ 0 & ; \text{otherwise} \end{cases}$$

6. Determine the linear convolution of $x(n) = \sum_{k=0}^2 \delta(n-k)$ and $h(n) = 2^n [u(n) - u(n-3)]$.

$$\therefore \text{Given, } x(n) = \sum_{k=0}^2 \delta(n-k)$$

$$\therefore x(n) = \delta(n) + \delta(n-1) + \delta(n-2)$$

$$\therefore x(n) = \delta(n) + \delta(n-1) + \delta(n-2) = 1$$

$$x(1) = \delta(1) + \delta(0) + \delta(-1) = 1$$

$$x(2) = \delta(2) + \delta(1) + \delta(0) = 1$$

$$\text{And, } h(n) = 2^n [u(n) - u(n-3)]$$

$$\text{Here, } u(n) = \{1, 1, 1, 1, \dots\}$$

$$u(n-3) = \{0, 0, 0, 1, 1, \dots\}$$

$$\text{So, } u(n) - u(n-3) = \{1, 1, 1\}$$

$$\therefore h(n) = 2^n \{1, 1, 1\} = \{1 \times 2^0 + 1 \times 2^1 + 1 \times 2^2\} = \{1, 2, 4\}$$

Now,

$$\text{Length of } x(n) = L = 3$$

$$\text{Length of } h(n) = M = 3$$

$$\text{Length of } y(n) = L+M-1 = 3+3-1 = 5.$$

$$\text{And, upper limit of } y(n) = 2+2 = 4.$$

$$\text{Lower limit of } y(n) = 0+0 = 0.$$

The convolution of $x(n)$ & $h(n)$ is given by,

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k) \cdot h(n-k)$$

$$y(n) = x(n) \cdot h(n) + x(n) \cdot h(n-1) + x(n-2) + x(n-3) + x(n-4)$$

At $n=0$: $y(0) = x(0) \cdot h(0) = 1$.

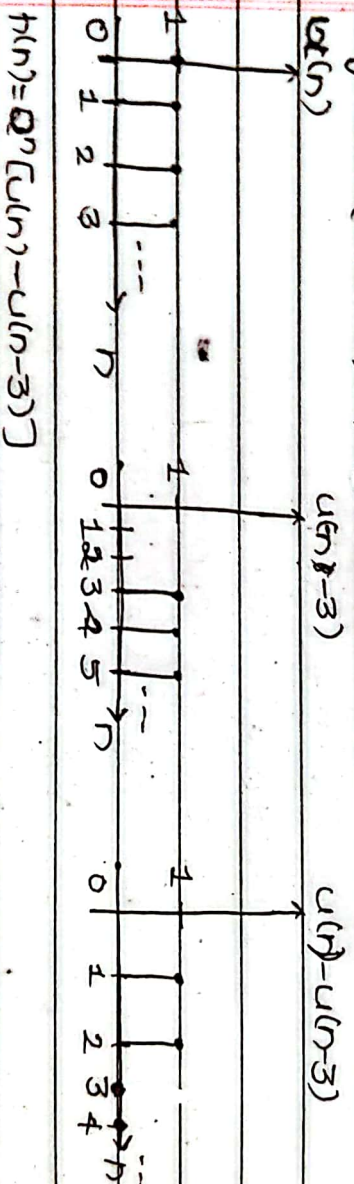
At $n=1$: $y(1) = x(0) \cdot h(1) + x(0) \cdot h(0) = 2 + 1 \times 1 = 3$

At $n=2$: $y(2) = x(0) \cdot h(2) + x(1) \cdot h(1) + x(0) \cdot h(0) = 4 + 2 + 1 = 7$.

At $n=3$: $y(3) = x(0) \cdot h(3) + x(1) \cdot h(2) + x(2) \cdot h(1) + x(0) \cdot h(0) = 4 + 2 + 0 = 6$.

At $n=4$: $y(4) = x(0) \cdot h(4) + x(1) \cdot h(3) + x(2) \cdot h(2) + x(3) \cdot h(1) + x(0) \cdot h(0) = 4$.

$$\therefore y(n) = \{1, 3, 7, 6, 4\}$$



7. A DT-signal is applied to a D.T. LTI system with unit impulse response $h(n)$. Find op or resp-